

A sharp uniform noise bound for interpolating oblique reconstructions

Candidate full solution to the operator-norm question in arXiv:1706.06444

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Result. If the sampling frame has lower frame bound A and $c = \cos(\varphi_{\mathcal{T}, \mathcal{U}}) > 0$, then every member of the source's interpolating family satisfies

$$\|Q_\lambda\|_{\text{op}} \leq \frac{1}{\sqrt{A}c}, \quad 0 \leq \lambda \leq 1.$$

This is the best possible universal bound for every fixed choice of $0 < A \leq B$, $0 < c \leq 1$, and λ : equality is attained simultaneously for all λ by an explicit three-dimensional example.

1 Source question and notation

Berger, Gröchenig, and Matz [1] study a frame (u_j) for a closed subspace $\mathcal{U} \subset \mathcal{H}$. Write $U : \ell^2 \rightarrow \mathcal{H}$ for its synthesis operator, U^* for its analysis operator, $S = UU^*$ for its frame operator, and $G = U^*U$ for its Gramian. Let A, B be frame bounds, so that

$$\text{spec}(S|_{\mathcal{U}}) \subset [A, B], \quad 0 < A \leq B < \infty.$$

For a closed reconstruction space $\mathcal{T}$, assume

$$c := \cos(\varphi_{\mathcal{T}, \mathcal{U}}) = \inf_{\substack{t \in \mathcal{T} \\ \|t\|=1}} \|P_{\mathcal{U}}t\| > 0.$$

For $0 < \lambda \leq 1$, put

$$\Sigma_\lambda = \lambda I + (1 - \lambda)G, \quad Q_\lambda d = \operatorname{argmin}_{g \in \mathcal{T}} \|\Sigma_\lambda^{-1/2}(U^*g - d)\|_2.$$

At $\lambda = 0$ the source uses $G^{\dagger/2}$ in place of $\Sigma_\lambda^{-1/2}$. These operators are perfect: $Q_\lambda U^*t = t$ for $t \in \mathcal{T}$.

Lemma 3.8 of the source bounds the different quantity $\|Q_\lambda U^*\|$. Immediately afterward, on source PDF page 16, the authors state that they do not know how to obtain a meaningful bound on $\|Q_\lambda\|$ itself. The relevant passage is reproduced below.

2 Proof intuition

The apparent obstruction is that $\|\Sigma_\lambda^{-1/2}\|$ blows up as $\lambda \downarrow 0$ whenever the sampling frame is redundant. But Q_λ annihilates the orthogonal complement of $\text{ran } U^*$, exactly where that blow-up occurs. On $\text{ran } U^*$ the preconditioner has norm $(\lambda + (1 - \lambda)A)^{-1/2}$.

The same preconditioner improves the lower bound of the weighted analysis map on $\mathcal{T}$ by the reciprocal factor $\sqrt{A/(\lambda + (1 - \lambda)A)}$. Taking its Moore–Penrose inverse and multiplying the two estimates makes all λ -dependence cancel. A one-dimensional reconstruction space aligned with a lower-frame-bound eigenvector makes both estimates equalities, proving optimality.

Thus $C_1 = 0$ on $\mathcal{R}(U^*)^\perp$, which is (33).

For showing $C_1 = C_3$ and (34) we repeat the proof of Theorem 3.3 verbatim. $\square$

The following Lemma gives a useful upper bound on the quasi-optimality constant of the operators Q_λ .

Lemma 3.8. *Let $\cos(\varphi_{\mathcal{T}, \mathcal{U}}) > 0$ and let Q_λ be defined as in Theorem 3.7. Then the quasi-optimality constant $\mu(Q_\lambda) = \|Q_\lambda U^*\|_{\text{op}}$ is bounded by*

$$\|Q_\lambda U^*\|_{\text{op}} \leq \frac{1}{\cos(\varphi_{\mathcal{T}, \mathcal{U}})} \sqrt{\frac{B(\lambda + A(1 - \lambda))}{A(\lambda + B(1 - \lambda))}}. \quad (35)$$

Proof. By Lemma 3.6 $(u_{j,\lambda})_{j \in \mathbb{N}}$ is a frame for $\mathcal{U}$ with frame bounds $\frac{A}{\lambda + (1 - \lambda)A}$ and $\frac{B}{\lambda + (1 - \lambda)B}$, and synthesis operator $L_\lambda = \Sigma_\lambda^{-\frac{1}{2}} U^*$. Using (18) for the frame $(u_{j,\lambda})_{j \in \mathbb{N}}$ we infer (35). $\square$

We observe that the upper bound in (35) is decreasing for $\lambda \rightarrow 0$, and for $\lambda = 0$ the upper bound coincides with the operator norm $\|Q_0 U^*\|_{\text{op}} = \frac{1}{\cos(\varphi_{\mathcal{T}, \mathcal{U}})}$. Unfortunately we do not know yet how to obtain a meaningful bound on the operator norm of Q_λ .

Remark 3.9. In [12] the authors consider a regularization term (Tikhonov regularization). The reconstruction operators corresponding to such a regularized least squares fit do not fulfill $Q(\langle f, u_j \rangle_{\mathcal{H}})_{j \in \mathbb{N}} = f$ for $f \in \mathcal{T}$ (see [12] equation (7)), and therefore do not belong to the class of reconstruction operators analyzed in this paper.

3.4. Numerical calculation of the coefficients. We now discuss how to calcu-

Figure 1: The extracted request after Lemma 3.8 in the source PDF, page 16.

3 The sharp theorem

Let $J : \ell^2(I) \rightarrow \mathcal{T}$ be a unitary synthesis map for an orthonormal basis of $\mathcal{T}$. Define

$$W_\lambda = \begin{cases} (\lambda I + (1 - \lambda)G)^{-1/2}, & 0 < \lambda \leq 1, \\ G^{\dagger/2}, & \lambda = 0, \end{cases} \quad D_\lambda = W_\lambda U^* J.$$

Lemma 1 (Two compensating estimates). *For $d_\lambda = \lambda + (1 - \lambda)A$,*

$$\|D_\lambda x\| \geq c \sqrt{\frac{A}{d_\lambda}} \|x\|, \quad \|W_\lambda|_{\text{ran } U^*}\| \leq \frac{1}{\sqrt{d_\lambda}}.$$

Both statements include $\lambda = 0$ with $d_0 = A$.

Proof. Because (u_j) is a frame for $\mathcal{U}$, $\text{ran } U^*$ is closed. Under the polar decomposition of $U^*|_{\mathcal{U}}$, the restriction of G to $\text{ran } U^*$ is unitarily equivalent to $S|_{\mathcal{U}}$. Hence

$$\text{spec}(G|_{\text{ran } U^*}) \subset [A, B],$$

and functional calculus gives the second estimate.

The same polar decomposition, or the standard push-through identity for frame operators, gives for $t \in \mathcal{T}$

$$\|W_\lambda U^* t\|^2 = \left\langle P_{\mathcal{U}} t, S(\lambda I + (1 - \lambda)S)^{-1} P_{\mathcal{U}} t \right\rangle.$$

The scalar function $s \mapsto s/(\lambda + (1 - \lambda)s)$ is increasing on $(0, \infty)$ (constant when $\lambda = 0$). Spectral calculus and the angle condition therefore yield

$$\|W_\lambda U^* t\|^2 \geq \frac{A}{d_\lambda} \|P_{\mathcal{U}} t\|^2 \geq \frac{A}{d_\lambda} c^2 \|t\|^2.$$

Taking $t = Jx$ proves the first estimate. At $\lambda = 0$ the middle operator is $P_{\mathcal{U}}$, so the same formula remains valid. $\square$

Theorem 1 (Optimal universal noise bound). *For every $0 \leq \lambda \leq 1$,*

$$\|Q_\lambda\|_{\text{op}} \leq \frac{1}{\sqrt{A}c}.$$

Moreover this constant is optimal even if A, B, c , and λ are all fixed in advance.

Proof. The representation theorem for Q_λ in the source, applied to an orthonormal basis of $\mathcal{T}$, reads

$$Q_\lambda = J D_\lambda^\dagger W_\lambda.$$

Here D_λ has closed range by Lemma 1. Its range is contained in $\text{ran } U^*$, so $D_\lambda^\dagger$ annihilates $(\text{ran } U^*)^\perp$. Since W_λ preserves the orthogonal decomposition $\ell^2 = \text{ran } U^* \oplus (\text{ran } U^*)^\perp$, Lemma 1 gives

$$\begin{aligned} \|Q_\lambda\| &\leq \|D_\lambda^\dagger\| \|W_\lambda|_{\text{ran } U^*}\| \\ &\leq \frac{1}{c} \sqrt{\frac{d_\lambda}{A}} \frac{1}{\sqrt{d_\lambda}} = \frac{1}{\sqrt{A}c}. \end{aligned}$$

For sharpness, fix arbitrary $0 < A \leq B$ and $0 < c \leq 1$. In $\mathcal{H} = \mathbb{C}^3$ with orthonormal basis e_1, e_2, e_3 , take

$$\mathcal{U} = \text{span}\{e_1, e_2\}, \quad u_1 = \sqrt{A}e_1, \quad u_2 = \sqrt{B}e_2, \quad \mathcal{T} = \text{span}\{t\}, \quad t = ce_1 + \sqrt{1 - c^2}e_3.$$

The sampling frame has optimal bounds A, B , and the subspace angle has cosine c . If $d = (d_1, d_2)$ and $g = \alpha t$, the weighted least-squares functional is

$$\frac{|\sqrt{A}c\alpha - d_1|^2}{\lambda + (1 - \lambda)A} + \frac{|d_2|^2}{\lambda + (1 - \lambda)B}.$$

Its unique minimizer is $\alpha = d_1/(\sqrt{A}c)$, including the Moore–Penrose endpoint $\lambda = 0$. Thus

$$Q_\lambda(d_1, d_2) = \frac{d_1}{\sqrt{A}c} t, \quad \|Q_\lambda\| = \frac{1}{\sqrt{A}c}$$

for every $\lambda \in [0, 1]$. This attains the asserted bound with all parameters fixed. $\square$

Corollary 1 (Complete universal bracket). *Put*

$$a_{\mathcal{T}} := \inf_{\substack{t \in \mathcal{T} \\ \|t\|=1}} \|U^* t\| > 0.$$

Then

$$\frac{1}{a_{\mathcal{T}}} = \|Q_1\| \leq \|Q_\lambda\| \leq \frac{1}{\sqrt{A}c}, \quad 0 \leq \lambda \leq 1.$$

Both inequalities can be equalities simultaneously.

Proof. $Q_1 = (U^*|_{\mathcal{T}})^\dagger$ is the Moore–Penrose left inverse, so its norm is $1/a_{\mathcal{T}}$. Every Q_λ is a left inverse of $U^*|_{\mathcal{T}}$; testing on unit vectors approaching the lower modulus $a_{\mathcal{T}}$ shows that no left inverse has smaller norm. The upper estimate is Theorem 1, and the sharpness example has $a_{\mathcal{T}} = \sqrt{A}c$. $\square$

4 Verification, novelty, and scope

The included script constructs random finite frames, redundant analysis operators, reconstruction subspaces, and all endpoint/interior values of λ . It checks perfect reconstruction and the theorem’s inequality. It separately verifies the equality construction for several prescribed values of A, B, c . These computations are sanity checks only; the proof is the functional-calculus argument above.

A bounded novelty search on 17 August 2026 covered the run’s four lightweight indexes and web searches for the exact title together with the phrase *operator norm* Q_λ , the source’s phrase *meaningful bound*, and weighted pseudoinverse norm bounds. The exact hits were the arXiv and published versions of the source; no later paper explicitly resolving this request was found. Novelty is therefore plausible, not certified.

The result completely determines the best universal bound in terms of the source’s frame bounds and subspace angle. It does not claim an exact formula for a specified frame, nor does it prove or disprove monotonicity of $\lambda \mapsto \|Q_\lambda\|$ for every individual frame. Human review should focus on the range restriction before the norm estimate and the $\lambda = 0$ Moore–Penrose endpoint. Verdict: **candidate full solution, likely valid**.

References

- [1] P. Berger, K. Gröchenig, and G. Matz, *Sampling and Reconstruction in Distinct Subspaces Using Oblique Projections*, arXiv:1706.06444; J. Fourier Anal. Appl. 25 (2019), 1080–1112.